

Codeword Polynomial

$$c(x) = c_0 + c_1x + c_2x^2 + \dots + c_{n-1}x^{n-1}$$

Each root β_i generates

$\updownarrow$ a row in H

$$\mathbf{c} = (c_0, c_1, c_2, \dots, c_{n-1})$$

$$c(x) \in \langle g(x) \rangle$$

implies

$$c(\beta_i) = 0$$

Geometric Locus

Roots of $g(x)$:

$$Z(g) = \{\beta_1, \beta_2, \dots, \beta_k\}$$

$$\underbrace{\begin{pmatrix} 1 & \beta_1 & \beta_1^2 & \dots & \beta_1^{n-1} \\ 1 & \beta_2 & \beta_2^2 & \dots & \beta_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \beta_k & \beta_k^2 & \dots & \beta_k^{n-1} \end{pmatrix}}_{\text{Parity-Check Matrix } H} \underbrace{\begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ \vdots \\ c_{n-1} \end{pmatrix}}_{\mathbf{c}^T} = \underbrace{\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{\mathbf{0}}$$

$$H\mathbf{c}^T = \mathbf{0} \iff c(x) \in \langle g(x) \rangle$$